GCE A LEVEL

A520U10-1

PREDICTED PAPER – ADVANCED

ECONOMICS – A level component 1

Economic Principles

1 hour 30 minutes

For Examiner's use only

Question	Maximum Mark	Mark Awarded
1-20	20	
21	3	
22	6	
23	3	
24	4	
25	8	
26	8	
27	8	
Total	**60**	

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Answer all questions.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question. A calculator may be used.

SECTION A

For each question in Section A, write the letter (A, B, C, D or E) that corresponds to your answer.

1. According to the Solow growth model, sustained long-run growth in GDP per capita ultimately depends on: **[1]**

- **A** Growth in the labour force
- **B** Growth in the capital stock
- **C** Exogenous technological progress (total factor productivity)
- **D** Increases in the savings rate
- **E** Higher rates of population growth

2. In a market with a negative externality of production, if the government sets a per-unit tax exactly equal to the marginal external cost (a Pigouvian tax), the post-tax outcome will: **[1]**

- **A** Eliminate all pollution
- **B** Internalise the externality and achieve the socially optimal output
- **C** Create a deadweight loss equal to the original welfare loss
- **D** Reduce output below the socially optimal level
- **E** Increase welfare by the full amount of tax revenue

3. A natural monopoly is characterised by an LRAC curve that falls continuously over the relevant range of output. If such a firm is regulated to price at marginal cost (P = MC), it will: **[1]**

- **A** Earn normal profits
- **B** Earn supernormal profits
- **C** Make a loss because MC is below AC
- **D** Be productively efficient but not allocatively efficient
- **E** Face zero deadweight loss and positive economic profit

4. In a market, $Q_d = 600 - 4P$ and $Q_s = -100 + 6P$ (both in units/day, P in £). If the government sets a maximum price of £50, the resulting market imbalance will be: **[1]**

- **A** An excess supply of 200 units
- **B** An excess demand of 200 units
- **C** An excess demand of 100 units
- **D** An excess supply of 100 units
- **E** Market equilibrium (no imbalance)

5. The UK has an inflation rate of 6% while the Eurozone has an inflation rate of 2%. Over the same period, the nominal £/€ exchange rate appreciates by 1%. The change in the UK's real effective exchange rate (against the Euro) is approximately: **[1]**

- **A** Appreciation of 3%
- **B** Appreciation of 5%
- **C** Depreciation of 3%
- **D** Depreciation of 5%
- **E** No change

6. The long-run Phillips curve is vertical at the Non-Accelerating Inflation Rate of Unemployment (NAIRU) because: **[1]**

- **A** All unemployment is voluntary
- **B** Workers adjust their inflation expectations to actual inflation in the long run
- **C** Money is neutral in both short and long run

D Monetary policy has no effect on inflation

E Supply shocks cancel out in the long run

7. The poorest 20% of households in Country X receive 5% of total income; the richest 20% receive 50%. A progressive tax-and-benefit system raises the share of the poorest 20% to 8% and reduces the richest 20% to 42%. The Gini coefficient will: **[1]**

A Rise

B Fall

C Remain unchanged

D Approach 1

E Become negative

8. Demand: $P = 120 - 2Q$. Supply: $P = 20 + 3Q$. The government imposes a specific tax of £10 per unit on producers. The loss in consumer surplus due to the tax is: **[1]**

A £76

B £140

C £176

D £380

E £400

9. In a perfectly contestable market, which of the following outcomes is most likely? **[1]**

A Firms will charge monopoly prices

B Firms will earn only normal profits in the long run, even with few competitors

C Firms will always operate below MES

D Firms will face inelastic demand

E Firms will engage in tacit collusion

10. The 'principal-agent problem' in the context of publicly listed companies refers to the conflict between: **[1]**

A Buyers and sellers

B Shareholders (principals) and managers (agents) arising from information asymmetry

C Workers and trade unions

D Government regulators and firms

E Creditors and debtors

11. At the zero lower bound on nominal interest rates, conventional monetary policy becomes ineffective because: **[1]**

A Fiscal policy always dominates

B Nominal rates cannot be cut further to stimulate demand

C Inflation is automatically zero

D Households save more when rates are low

E Banks hoard all reserves

12. In a closed economy, the theoretical multiplier is $k = 1/(1-MPC)$. Empirical estimates of the multiplier in open economies with flexible exchange rates and partial crowding out are typically: **[1]**

A Larger than the theoretical value

B About equal to the theoretical value

C Smaller than the theoretical value

D Negative

E Infinite

13. A government wishes to maximise tax revenue from a specific indirect tax. This is most likely to be achieved when demand for the taxed product is: **[1]**

- **A** Perfectly elastic
- **B** Price elastic
- **C** Perfectly inelastic
- **D** Unit elastic
- **E** Price inelastic with many complements

14. Country A experiences a 10% improvement in its terms of trade. Its trade volume (quantity of exports and imports) remains unchanged. The most likely effect on its current account balance is: **[1]**

- **A** The current account will deteriorate
- **B** The current account will improve
- **C** The current account will be unchanged
- **D** The current account will shift to surplus only if PEDs are elastic
- **E** Cannot be determined without more information

15. Quantitative easing (QE) is most likely to stimulate AD through all of the following channels EXCEPT: **[1]**

- **A** Lower long-term interest rates (portfolio rebalancing)
- **B** Higher asset prices and wealth effects
- **C** A weaker exchange rate
- **D** Increased bank lending to SMEs in every case
- **E** Signalling commitment to low rates for longer

16. The use of 'nudges' (e.g. default enrolment into pension schemes) in public policy is based on the behavioural insight that: **[1]**

- **A** Consumers always make fully rational choices
- **B** Small changes in choice architecture can significantly alter behaviour
- **C** Economic agents have identical preferences
- **D** Market failures do not exist in practice
- **E** Information provision alone solves all market failures

17. The 'tragedy of the commons' is most commonly associated with which pair of characteristics? **[1]**

- **A** Non-rival and excludable
- **B** Rival and excludable
- **C** Non-rival and non-excludable
- **D** Rival and non-excludable
- **E** Non-rival and non-diminishable

18. In year 1, nominal GDP is £2,000bn and the GDP deflator is 100. In year 2, nominal GDP rises to £2,184bn and the GDP deflator is 104. Real GDP growth (rounded) is: **[1]**

- **A** 2.0%
- **B** 4.0%
- **C** 5.0%
- **D** 9.2%
- **E** 13.2%

19. The Kuznets curve hypothesises that, as a country develops, income inequality: **[1]**

- **A** Continuously rises
- **B** Continuously falls
- **C** First rises, then falls (inverted-U shape)
- **D** First falls, then rises
- **E** Remains constant

20. The demand for a product is given by Q = 500 − 10P and supply by Q = 20P − 100. The government introduces a specific tax of £6/unit paid by producers. The new equilibrium quantity is: **[1]**

- **A** 220
- **B** 240
- **C** 260
- **D** 280
- **E** 300

SECTION B

Answer all the questions in the spaces provided.

21. The diagram below shows the kinked demand curve faced by an oligopoly firm, with the discontinuous 'gap' in the MR curve between points A and B. **[3]**

[Diagram: kinked demand curve with an elastic upper segment and inelastic lower segment, kink at the current price P*. MR curve has a vertical discontinuity at Q*]

Using the diagram, explain why price stability is a common feature of oligopoly markets.

22. The diagram below shows an economy with a Keynesian AS curve (perfectly elastic at low output, becoming vertical at full employment). The economy is currently in a deep recession at Y_1. **[6]**

[Diagram: Keynesian AS curve (L-shaped / three-stage: flat, upward-sloping, vertical), with AD intersecting in the flat portion at Y_1]

Consider the extent to which a £50bn expansionary fiscal stimulus would be an effective policy response. Adapt the diagram as part of your answer.

23. Lithium is a key input in electric vehicle batteries. Global demand is forecast to quadruple over the next decade. Lithium mines take 7–10 years to develop from initial exploration to full production. Existing mines are operating at near-full capacity. **[3]**

With the aid of a diagram, comment on the likely consequences for lithium prices over the next 5 years, referring to the price elasticity of supply.

24. A firm operates in two distinct stages of production: a small-scale 'artisan' stage (1–100 units) using skilled labour, and a large-scale 'factory' stage (100+ units) requiring substantial investment in machinery. The firm's short-run total cost curve has a distinctive kink at 100 units, with TC jumping from £5,000 at Q=99 to £25,000 at Q=100. **[4]**

Explain the shape of this firm's short-run total cost curve and the economic significance of the discontinuity.

25. The pay-off matrix below shows the expected changes in real GDP (%, annualised) for two economies depending on their trade policy stance. The economies are significant trading partners. **[8]**

	Economy Y: Free Trade	Economy Y: Retaliatory Tariffs
Economy X: Free Trade	+2.4%, +3.0%	-1.8%, +1.5%
Economy X: Tariffs on Y's exports	+1.2%, -2.0%	-3.5%, -4.0%

Additional context:

• Economy X is a large developed economy; Economy Y is a fast-growing emerging economy.

• The WTO average 'bound' tariff for developed economies is approximately 3.5%; for developing economies, around 15%.

• Recent empirical studies estimate trade wars reduce global GDP by 0.8–1.5%.

With reference to the matrix and economic theory, discuss whether the optimal trade policy is unilateral free trade.

26. The UK has a legally binding target of reaching 'net zero' greenhouse gas emissions by 2050. Read the information below. **[8]**

The UK's current approach combines:

• The UK Emissions Trading Scheme (ETS): a 'cap-and-trade' system covering ~30% of emissions, with the carbon price averaging £40/tonne in 2024 (down from £90/tonne in 2023).

• Fuel duty on petrol and diesel (frozen in nominal terms since 2011).

• Subsidies for renewables (via Contracts for Difference).

• Various regulations and standards.

Key data:

• UK CO_2 emissions have fallen by 48% since 1990.

• Renewable share of UK electricity: 43% (2023), up from 7% in 2010.

• UK ETS revenue: £5.5bn (2023), £3.2bn (2024).

• Climate Change Committee estimates the shadow cost of carbon for policy should be £240/tonne by 2030.

• Global CO_2 emissions continue to rise; UK accounts for ~1% of world emissions.

• Price Elasticity of Demand for petrol: ≈ −0.3 (short run), ≈ −0.7 (long run).

With reference to the data above, consider the extent to which the UK's carbon pricing policies are effective at correcting the market failure of climate change.

27. The table below presents development indicators for selected economies, comparing 'extractive' vs 'inclusive' institutional frameworks (as theorised by Acemoglu and Robinson). **[8]**

Country	GDP per capita ($, PPP)	Rule of Law Index (0-1, higher=better)	HDI	Corruption Perceptions rank /180	Institutional classification
South Korea	52,000	0.81	0.925	32	Inclusive
North Korea	1,700	0.14	<0.5 (est.)	172	Extractive
Chile	28,000	0.67	0.855	29	Mixed (largely inclusive)
Argentina	26,500	0.52	0.842	98	Mixed/extractive
Botswana	17,100	0.65	0.693	45	Inclusive (for region)
Zimbabwe	2,800	0.36	0.593	149	Extractive
Singapore	116,000	0.80	0.938	5	Inclusive
Nigeria	5,900	0.44	0.535	145	Mixed/extractive

Using the information above and relevant economic theory, discuss the extent to which institutional quality is the most fundamental determinant of long-run economic development.

END OF PAPER